



AI Infrastructure Architecture Evolution

2015-2026

Xeon · GPU · Gaudi · Ethernet · IPU
Packaging · Photonics · FPGA · Systems · Software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Executive synthesis and scope

Intel's AI infrastructure history is a history of several interacting architectures, rather than a single accelerator succession. Xeon remains the general-purpose execution and orchestration layer. Gaudi builds a training system around matrix engines and integrated Ethernet. Ponte Vecchio pursues heterogeneous packaging and high-performance scientific computing. IPU's remove infrastructure work from tenant CPUs. The latest inference GPU direction emphasizes model capacity and deployment into air-cooled servers. The central architectural question is how these independently developed components form a useful system.

Coverage runs from 2015 to 28 September 2026, with a short Haswell baseline. The organizing unit is the product generation. CPU and accelerator sections receive the greatest depth; networking, IPU's, packaging, optics, FPGA, memory, software and systems complete the portfolio. Client and edge products appear where they explain shared technology or a distinct deployment model. This is a curated technical history, not an inventory of every SKU or every Intel-affiliated conference paper.

The most consequential CPU transitions are the move from ring to mesh, the separation of private-cache and last-level-cache roles, the arrival of AMX and integrated offload engines, and the split into performance-core and efficiency-core server products. Chiplets then change from a way to extend die area into a way to place cores, cache and I/O on different processes. Clearwater Forest makes that separation vertical; Diamond Rapids extends it into a new announced performance-core platform. On accelerators, continuity is weaker: technical achievement and commercial roadmap survival must be assessed separately.

Portfolio map at the cutoff

Layer	Families	Architectural role	Status boundary
Host compute	Xeon Scalable / Xeon 6 / 6+	General compute, AMX, orchestration	Diamond Rapids disclosed; GA not established
Accelerators	Gaudi / GPU Max / Flex / Crescent Island	Training, HPC, media and inference	Crescent Island announced; Falcon Shores internal only
Network / infrastructure	E810 / E830 / E835; IPU E2000/E2100	RDMA, packet processing, storage isolation	NIC and IPU are different device classes
Adjacent technology	EMIB / Foveros / OCI / Altera / Optane	Integration, optical links, programmable logic, memory	OCI demonstration; Altera 49% retained; Optane exited

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

02 How to read the numbers

All bandwidths use decimal GB/s or TB/s. DRAM peak equals channels × transfer rate × eight data bytes; ECC bits do not add payload bandwidth. PCIe figures are per direction and precede protocol overhead. Ethernet Gb/s is divided by eight before comparison with memory bandwidth. A bidirectional aggregate is divided by two only when the source makes that convention explicit. HBM, package fabric, scale-up and scale-out bandwidth remain separate budgets.

CPU rows use representative high-end shipping configurations, not a composite assembled from different SKUs. Accelerator arithmetic is dense peak, with multiply-add counted as two operations. Tensor BF16 is not vector BF16; emulated formats are not native formats. Historical shipping does not mean currently orderable. Announcement, sampling and general availability are distinct. n/d means the reviewed evidence did not establish the value; n/a means the field does not apply. Sources are collected in the final source map and bibliography so the narrative can remain readable.

03 Xeon generation atlas

The table is a reference-SKU comparison: 8592+ is 350 W, while other Emerald Rapids SKUs have different power limits. Cooper Lake and Ice Lake are both called third-generation Xeon Scalable but address different socket segments and use different cores. Sierra Forest’s shipping 144-core reference must not be silently replaced by an earlier 288-core roadmap target. The Diamond Rapids row records the August 2026 disclosure, not final SKU specifications. Its memory figure is a channel-rate calculation; the reviewed announcement does not establish the final DIMM qualification matrix.

CPU reference configurations · 1/3

Generation / reference SKU	Codename	GA	Status	Core march	Cores / threads
E5 v4 / 2699 v4	Broadwell-EP	2016	Historical shipping	Broadwell	22 / 44
1st / 8180	Skylake-SP	2017	Historical shipping	Skylake	28 / 56
2nd / 8280	Cascade Lake	2019	Historical shipping	Skylake-derived	28 / 56
3rd / 8380HL	Cooper Lake	2020	Historical shipping	Skylake-derived	28 / 56
3rd / 8380	Ice Lake-SP	2021	Historical shipping	Sunny Cove	40 / 80
4th / 8490H	Sapphire Rapids	2023	Historical shipping	Golden Cove	60 / 120
Xeon Max / 9480	Sapphire Rapids HBM	2023	Historical shipping	Golden Cove	56 / 112
5th / 8592+	Emerald Rapids	2023-12	Historical shipping	Raptor Cove	64 / 128
6 E / 6780E	Sierra Forest	2024-06	Shipping	Crestmont	144 / 144
6 P / 6980P	Granite Rapids	2024-09	Shipping	Redwood Cove	128 / 256
6+ / 6990E+	Clearwater Forest	2026-06	Shipping	Darkmont	288 / 288
Next Xeon	Diamond Rapids	n/d	Disclosed; GA n/d	New performance core	256 / n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CPU reference configurations · 2/3

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
E5 v4 / 2699 v4	14 nm	Monolithic; ring	256 KB	55 MB inclusive
1st / 8180	14 nm	Monolithic; mesh	1 MB	38.5 MB non-inclusive
2nd / 8280	14 nm	Monolithic; mesh	1 MB	38.5 MB
3rd / 8380HL	14 nm	Monolithic; scale-up	1 MB	38.5 MB
3rd / 8380	10 nm	Monolithic; mesh	1.25 MB	60 MB
4th / 8490H	Intel 7	4 compute tiles; EMIB	2 MB	112.5 MB
Xeon Max / 9480	Intel 7	4 tiles + 4 HBM stacks; EMIB	2 MB	112.5 MB
5th / 8592+	Intel 7	2 tiles high-end; EMIB	2 MB	320 MB
6 E / 6780E	Intel 3 + Intel 7 I/O	Compute + 2 I/O; EMIB	4 MB / 4-core cluster	108 MB
6 P / 6980P	Intel 3 + Intel 7 I/O	3 compute + 2 I/O; EMIB	2 MB	504 MB
6+ / 6990E+	Intel 18A compute; Intel 3 base; Intel 7 I/O	12 compute + 3 base + 2 I/O; 3D + EMIB	n/d	576 MB
Next Xeon	Intel 18A-P	Foveros Direct 3D; UCLe-S	n/d	1.28 GB

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

CPU reference configurations - 3/3

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
E5 v4 / 2699 v4	DDR4; 4; 2400; 76.8	2S; QPI 9.6 GT/s	40 PCIe 3; CXL n/a	145 W	AVX2
1st / 8180	DDR4; 6; 2666; ~128	2/4/8S; 3 UPI 10.4 GT/s	48 PCIe 3; CXL n/a	205 W	AVX-512
2nd / 8280	DDR4; 6; 2933; ~140.8	2/4/8S; 3 UPI 10.4 GT/s	48 PCIe 3; CXL n/a	205 W	AVX-512; VNNI
3rd / 8380HL	DDR4; 6; 3200; 153.6	4/8S; 6 UPI 10.4 GT/s	48 PCIe 3; CXL n/a	250 W	AVX-512; BF16; VNNI
3rd / 8380	DDR4; 8; 3200; 204.8	2S; 3 UPI 11.2 GT/s	64 PCIe 4; CXL n/a	270 W	AVX-512; VNNI
4th / 8490H	DDR5; 8; 4800; 307.2	up to 8S; 4 UPI 16 GT/s	80 PCIe 5; CXL 1.1	350 W	AVX-512; AMX INT8/BF16
Xeon Max / 9480	DDR5; 8; 4800; 307.2 + HBM2e 64 GB ~1000	2S; UPI 16 GT/s	80 PCIe 5; CXL 1.1	350 W	AVX-512; AMX INT8/BF16
5th / 8592+	DDR5; 8; 5600; 358.4	2S; 4 UPI 20 GT/s	80 PCIe 5; CXL 1.1	350 W	AVX-512; AMX INT8/BF16
6 E / 6780E	DDR5; 8; 6400; 409.6	2S; UPI 24 GT/s	88 PCIe 5; CXL 2.0; shared lane budget	330 W	AVX2; VNNI; no AMX
6 P / 6980P	DDR5; 12; 6400; 614.4 / MRDIMM 8800; 844.8	2S; UPI 24 GT/s	96 PCIe 5; CXL 2.0	500 W	AVX-512; AMX INT8/BF16/F P16
6+ / 6990E+	DDR5; 12; 8000; 768	2S; link details n/d	96 PCIe 5; CXL allocation n/d	450 W	AVX2; no AMX
Next Xeon	16; 12800 MT/s; 1638.4 derived	n/d	128 PCIe 6; CXL 3.0	n/d	APX; enhanced AMX

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

04 Haswell and Broadwell: the monolithic baseline

Haswell-EP established the 2015 starting point with AVX2, DDR4 and a ring-connected server die. Broadwell-EP's 2016 shrink to 14 nm raised the high-end E5 core count to 22 while retaining a 256 KB private L2 and a large inclusive LLC. The important constraint is the balance among cores, four memory channels and the ring. More cores increase aggregate execution demand without proportionately increasing off-chip bandwidth. Even before dedicated AI hardware, streaming and sparse workloads could run out of memory bandwidth before they ran out of arithmetic.

E5-2600 and E7 should be kept separate: E7 emphasized larger coherent socket domains and reliability, while E5 defined mainstream one- and two-socket servers. Xeon D brought related x86 technology into an integrated, lower-power SoC for network and edge systems. Desktop Core shared a core lineage but not the server's memory-channel count, cache organization or socket interconnect. This distinction remains necessary throughout the report: core architecture is only one component of a server product.

05 Skylake, Cascade Lake and Cooper Lake

Skylake-SP's 2017 change is architectural, not merely a core-count increment. A two-dimensional mesh replaces the ring. Private L2 grows from 256 KB to 1 MB, while the LLC becomes smaller per core and non-inclusive. Capacity comparisons that add or compare L3 alone therefore misread the hierarchy. The larger private cache retains active data closer to execution; the LLC can act more effectively as a shared capacity and victim resource. Mesh routing, directory behavior and sub-NUMA clustering make placement important even inside one socket.

Six DDR4 channels, UPI and AVX-512 accompany the mesh. AVX-512 doubles vector width relative to AVX2, but supported execution resources and operating frequency depend on SKU and instruction mix. Sustained application gain cannot be inferred from width alone. Cascade Lake keeps this broad organization and adds VNNI to reduce the instruction work for low-precision dot products. Its Optane persistent-memory support changes capacity and persistence semantics, not simply memory speed. The 9200-series Cascade Lake AP places two dies in one package; its 56-core headline is a separate product construction.

Cooper Lake adds AVX-512 BF16 and stronger multi-socket connectivity for scale-up systems. It is not the same silicon generation as Ice Lake, despite sharing the third-generation Xeon Scalable label. BF16 keeps FP32-like exponent range while shortening the significand, making it useful for neural-network arithmetic with appropriate accumulation. This is still vector execution, before AMX's tile-based matrix datapath. Hot Chips' Cascade Lake presentation supplies the feature transition; product and optimization documentation identify the actual deployment and programming constraints.

06 Ice Lake: a wider core and a wider server

Ice Lake-SP arrives in 2021 with Sunny Cove, up to 40 cores, eight DDR4 channels and PCIe 4. The core expands out-of-order resources and the memory subsystem relative to the Skylake family, while the server implementation supplies a 1.25 MB L2 per core. A client Sunny Cove block diagram is useful background but is not a substitute for server cache and uncore documentation. The shift to eight memory channels raises theoretical DDR bandwidth to 204.8 GB/s, a substantial system change alongside the process and core improvements.

The server gains a more useful I/O budget for storage and accelerators, while SGX and memory-protection features address cloud isolation requirements. Yet it remains a monolithic CPU with a conventional external accelerator relationship. PCIe 4 x16 offers about 31.5 GB/s in one direction before higher-level overhead, far below the local memory bandwidth of contemporary HBM accelerators. Offloading a small kernel can therefore lose to keeping the computation on the CPU once transfer and synchronization are included. Xeon D derivatives adapt related technology to edge connectivity and long-lived platforms rather than reproducing the full server socket.

07 Sapphire Rapids and Xeon Max: tiles and engines

Sapphire Rapids, commercially launched in January 2023, combines Golden Cove cores with DDR5, PCIe 5, CXL 1.1 and a four-tile EMIB construction in its high-end configuration. Each tile participates in a package-wide coherent system rather than exposing four independent processors. The ISSCC 2022 disclosure describes a quasi-monolithic integration approach and 10 TB/s of total die-to-die bandwidth across 20 interfaces at 0.5 pJ/bit. Those are implementation-level aggregate figures; they are not socket DRAM bandwidth or a software-visible remote-memory guarantee. Intel 7 is the commercial process name despite differing terminology in some conference summaries.

AMX changes CPU AI execution more fundamentally than another vector-width increase. Software configures tile registers and invokes matrix operations on blocks of INT8 or BF16 data. Libraries must select supported kernels, pack operands and manage tile state. Matrix peak rises faster than DRAM bandwidth, so cache reuse and batching become more valuable. AMX is therefore especially useful when a workload can retain data in cache or amortize weight reads; it does not make every inference workload compute-bound. CPU computation remains attractive for control-heavy operators and modest deployment sizes.

DSA, IAA, QAT and DLB target different kinds of overhead: data movement, analytics/compression, cryptography/compression, and event distribution. Their value comes from removing work and cache pollution from cores, but queue submission, completion, data locality and minimum useful transfer sizes still matter. The ASPLOS 2024 DSA study measures these tradeoffs on actual hardware. A block appearing on the package is not evidence that an application automatically uses it; software enablement and SKU licensing/configuration must be checked.

Xeon Max adds 64 GB of HBM2e around the CPU tiles, exposing HBM-only, flat and cache usage modes. This brings roughly 1 TB/s of local high-bandwidth memory to CPU code without converting the application to a GPU execution model. In flat mode, allocation policy determines which objects receive HBM service; in cache mode, locality and working-set behavior determine the benefit. HBM and DDR peaks cannot simply be added as the sustained bandwidth of one stream. Capacity-limited applications may still require DDR, while bandwidth-heavy scientific kernels can benefit substantially from the near-memory tier.

08 Emerald Rapids: cache and integration efficiency

Emerald Rapids launches in December 2023 on the established platform, with up to 64 cores and 320 MB LLC. The high-end implementation uses two larger tiles rather than Sapphire Rapids' four-tile arrangement. Reducing tile boundaries changes fabric traffic, physical integration and cache organization without requiring a wholesale platform replacement. ISSCC 2024 is the useful implementation anchor: the disclosure emphasizes the Intel 7 multichip construction and substantially larger shared cache. Its reported iso-power workload gains are measured comparisons, not a universal IPC multiplier.

The representative 64-core part provides 5 MB of LLC per core, versus 1.875 MB for the 60-core Sapphire Rapids reference. That capacity increase can reduce expensive DRAM accesses for databases, graph structures and reused model data. DDR5-5600 also raises the channel peak, but the architectural gain depends on whether a workload was cache-capacity-limited, bandwidth-limited or compute-limited. This generation demonstrates why packaging and cache redesign can matter even when the process node and major matrix ISA remain unchanged.

09 Xeon 6: separating performance and density

Hot Chips 2023 disclosed the shared platform strategy behind Granite Rapids and Sierra Forest before their 2024 launches. Compute tiles and I/O tiles are separated, allowing Intel 3 cores to be combined with Intel 7 I/O. The server family offers common infrastructure with different compute choices; it is not a hybrid desktop CPU mixing P-cores and E-cores in one socket. DDR5, CXL 2.0 and built-in accelerators belong to the platform story, while vector capability, cache organization and thread behavior remain core-family-specific.

Granite Rapids' high-end 6900P configuration reaches 128 Redwood Cove cores, three compute tiles, twelve memory channels and 500 W on the 6980P reference. MRDIMM raises the qualified memory data rate to 8800 MT/s, yielding 844.8 GB/s of theoretical channel bandwidth. AMX adds FP16 support. The engineering point is balanced throughput: a much larger core complex needs more memory bandwidth and package connectivity, not just faster cores. The 6700P expansion in 2025 serves a different platform and channel envelope; its specifications should not be mixed with 6900P.

Sierra Forest's 144-core shipping reference uses Crestmont E-cores without SMT or AMX. Four-core clusters share L2. The intended gain is density for independent throughput-oriented services, not equivalence to 144 large vector cores. A cloud service with many small requests can exploit this configuration; a heavily vectorized AMX workload cannot assume the same instruction path as Granite Rapids. At full population, the reference has about 2.84 GB/s of DRAM channel bandwidth per core. Core count is therefore meaningful only alongside workload concurrency, cache behavior and memory demand.

10 Clearwater Forest and Diamond Rapids

Clearwater Forest was discussed at Hot Chips 2025 and MWC 2026, then launched as Xeon 6+ in June 2026. Its 288 Darkmont E-cores occupy twelve Intel 18A compute chiplets above three Intel 3 base dies, with Intel 7 I/O. Foveros Direct 3D and EMIB combine vertical and lateral integration. This separates dense compute from cache/fabric and I/O implementation. The architectural opportunity is process specialization and shorter dense vertical connections; the corresponding engineering burden includes assembly yield, thermal coupling, power delivery and fabric placement.

The 6990E+ reference has 576 MB LLC and twelve DDR5-8000 channels. Compared with Sierra Forest's reference, core count doubles while DRAM peak rises from 409.6 to 768 GB/s. Per-core channel bandwidth therefore changes from about 2.84 to 2.67 GB/s, while average LLC capacity rises from 0.75 to 2 MB per core. This is a cache-and-density design, not a claim that every core receives more external bandwidth. Its 288 cores are 288 hardware threads; multiplying by two would incorrectly apply P-core SMT behavior.

At Hot Chips 2026, Intel disclosed Diamond Rapids with up to 256 new cores, 1.28 GB LLC, sixteen memory channels at 12800 MT/s, 128 PCIe 6 lanes and CXL 3.0. Intel 18A-P, Foveros Direct 3D, UCle-S, a unified memory fabric, APX and enhanced AMX define the announced direction. The design moves performance-core Xeon toward a much larger distributed package. The public announcement establishes these headline parameters; it does not establish final SKU power, detailed queue sizes, thread count or commercial availability. Coral Rapids remains a subsequent roadmap name rather than a table of shipping specifications.

11 Accelerator generation atlas

Accelerator reference configurations · 1/4

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Xeon Phi 7290	Knights Landing	2016	Historical shipping	14 nm	CPU + on-package MCDRAM
Gaudi / HL-205	Gaudi	2021 AWS DL1	Historical shipping	16 nm	1 compute die + HBM
Gaudi 2	Gaudi 2	2022	Historical shipping	7 nm	1 compute die + HBM
Gaudi 3 / HL-325L	Gaudi 3	2024	Shipping	5 nm	2 compute dies + HBM
GPU Max 1550	Ponte Vecchio / Xe-HPC	2023	Historical shipping	TSMC N5 compute; Intel 7 base; TSMC N7 link	47-tile integration; EMIB + Foveros
GPU Flex 170	Xe-HPG	2022	Historical shipping	TSMC N6	Single GPU
Crescent Island	Xe3P	n/d	Disclosed; GA n/d	n/d	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Accelerator reference configurations - 2/4

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Xeon Phi 7290	72 cores / 288 threads	n/a	n/a	n/a	~3.46 TF / n/a
Gaudi / HL-205	8 TPC; MME	n/d	n/a	n/a	n/a
Gaudi 2	24 TPC / 2 MME	432 TF	865 TF	n/a	n/a
Gaudi 3 / HL-325L	64 TPC / 8 MME	1678 TF	1678 TF	n/a	n/a
GPU Max 1550	128 Xe cores / 1024 XMX	839 TF	n/a	n/a	52 TF native; split n/d
GPU Flex 170	32 Xe cores	n/d	n/a	n/a	n/d
Crescent Island	32 Xe cores / 256 XMX	n/d	n/d	n/d	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Accelerator reference configurations - 3/4

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM	Scale-up: links; GB/s per direction; domain
Xeon Phi 7290	MCDRAM; 8; 16; >0.4 measured STREAM; DDR4 separate	36 MB distributed L2	n/a; optional Omni-Path fabric
Gaudi / HL-205	HBM2; n/d; 32; ~1	n/d	10 x 100GbE; 125 total; 8-device server
Gaudi 2	HBM2e; 6; 96; 2.46	48 MB SRAM	24 x 100GbE; 300 total; 8-device server
Gaudi 3 / HL-325L	HBM2e; 8; 128; 3.7	96 MB SRAM	24 x 200GbE; 600 total; 8-device server
GPU Max 1550	HBM2e; 8; 128; 3.2	408 MB L2	Xe Link; 318 on Lenovo reference; 4-GPU domain
GPU Flex 170	GDDR6; n/a; 16; 0.576	n/d	n/a
Crescent Island	LPDDR5X; n/a; up to 480; n/d	n/d	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Accelerator reference configurations - 4/4

Product	Host link	Power / cooling	Form factor
Xeon Phi 7290	Self-hosted; PCIe 3	245 W	Socketed processor
Gaudi / HL-205	PCIe 4	n/d	OAM
Gaudi 2	PCIe 4 x16	600 W	OAM
Gaudi 3 / HL-325L	PCIe 5 x16	Up to 900 W; air/liquid variants	OAM
GPU Max 1550	PCIe 5 x16	600 W; platform-dependent	OAM
GPU Flex 170	PCIe 4 x16	150 W; passive air	PCIe
Crescent Island	PCIe; generation n/d	350 W; air	PCIe

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Gaudi 3 uses the retrieved architecture whitepaper’s 1678 TFLOPS BF16/FP8 values, not a mixture with other clock-point marketing figures. The 600 W HL-338 PCIe card is a separate configuration and should not inherit OAM peak numbers without verification. GPU Max’s 636 GB/s bidirectional Xe Link figure comes from the specific Lenovo system specification and is normalized to 318 GB/s per direction. It is not a claim that every Ponte Vecchio system exposes the same external link topology. Xeon Phi is included as a historical many-core accelerator lineage, although Knights Landing can boot its own operating system.

12 Xeon Phi and Nervana: earlier parallel-compute paths

Knights Landing, disclosed at Hot Chips 2015 and commercialized in 2016, combines many small x86 cores, four hardware threads per core, wide vector execution, a mesh and 16 GB of on-package MCDRAM. It is more than a PCIe coprocessor: a self-hosted system can run a conventional operating system and use DDR4 for capacity. Cache, flat and hybrid memory modes expose an early version of the placement problem later revisited by Xeon Max. The software appeal is x86 and familiar parallel programming; achieving throughput still requires vectorization, many concurrent threads and careful memory use.

Knights Mill redirected the many-core line toward deep learning with specialized vector instructions, but it did not become the AMX architecture used by later Xeons. Nervana's NNP-T and NNP-I subsequently explored dedicated training and inference datapaths. Intel's acquisition of Habana and the later concentration on Gaudi changed the commercial direction. These branches belong in the history because they reveal changing assumptions about programmability, numerical formats and interconnect, not because they form a continuous shipping sequence into current GPUs. A discontinued program's projected successor must not appear as a delivered generation.

13 Gaudi 1, 2 and 3: Ethernet inside the accelerator

Gaudi begins with a different decomposition from a graphics-derived GPU. Matrix multiplication engines perform the dense linear algebra; programmable Tensor Processor Cores handle the surrounding operations; local HBM supplies capacity and bandwidth; integrated RoCE-capable Ethernet ports carry device communication. AWS DL1's October 2021 general availability provides a concrete first-generation deployment milestone: eight Gaudi devices with 32 GB HBM each. Its 400 Gb/s instance networking is a cloud service limit and must not be confused with the sum of every on-chip Gaudi port.

Gaudi 2 scales the design to 24 TPCs, two MMEs, 96 GB HBM2e and 24 × 100 GbE ports. The matrix path supports BF16 and FP8 at different peak rates. A large software-managed SRAM helps reuse data and reduce HBM traffic, but it is not automatically equivalent to a CPU's coherent last-level cache. The architecture depends on compiler scheduling across matrix, vector, DMA and communication resources. Its main system distinction is the ability to build both the local accelerator group and the external fabric from Ethernet mechanisms.

Gaudi 3 moves to two compute dies, eight MMEs, 64 TPCs, 128 GB HBM2e and 24 × 200 GbE. The two-die construction expands compute and memory interfaces, while communication must cross internal partitions when data is not local. On the eight-accelerator baseboard, 21 ports per device supply three links to each of seven peers, leaving three ports for scale-out. That partitions the 600 GB/s one-way all-port budget into 525 GB/s local peer traffic and 75 GB/s external traffic. The total cannot be counted independently in both directions and both roles.

Integrated networking removes the need for a separate NIC for these accelerator links, but it does not remove the Ethernet switching fabric, congestion control, cabling or collective algorithms. HCCL, SynapseAI and framework integration determine whether the hardware overlaps communication with computation and supports the needed operators. Gaudi 3's PCIe variant extends deployment into more conventional servers with a different power envelope. The practical comparison with a GPU system is therefore a complete supported training or inference configuration, including software and network topology, rather than a single peak-FLOPS column.

14 Ponte Vecchio: a heterogeneous HPC package

Ponte Vecchio's ISSCC 2022 disclosure is one of the clearest examples of implementation driving architecture. The 47-tile integration includes sixteen 5 nm compute tiles and multiple process technologies, connected through Foveros and EMIB. The count is not 47 compute dies. Compute tiles, base/cache resources, link tiles and HBM perform different roles. This lets each function use a suitable manufacturing and integration choice, at the cost of complex assembly, testing, power delivery and thermal management. The conference prototype's measured figures and the later GPU Max SKU peaks are different evidence points.

The commercial Max 1550 combines 128 Xe-HPC cores, XMX matrix execution, strong native FP64, 128 GB HBM2e and a large distributed L2. Unlike an inference-only accelerator, it invests heavily in scientific arithmetic and HPC execution. Xe Link connects peer GPUs; PCIe connects the host. A unified programming abstraction does not make CPU DDR and GPU HBM physically uniform or equally fast. Aurora demonstrates a system built around CPU and GPU high-bandwidth memory, but its communication and placement costs still have to be managed by applications and runtimes.

Ponte Vecchio's technical success as an integration vehicle should be separated from the breadth and longevity of the commercial line. Lenovo's SD650-I V3 product guide is explicitly marked withdrawn, despite documenting a real four-GPU system. Falcon Shores was later designated an internal test chip rather than a marketed successor. Consequently, it is misleading to draw an uninterrupted arrow from Xeon Phi through GPU Max to a shipping Falcon Shores platform. Product continuity, software support and deployable system supply are architecture constraints in their own right.

15 Flex, Arc and Crescent Island: inference branches

Flex 140 and Flex 170 serve media processing, cloud graphics, virtualization and visual inference with Xe-HPG technology and GDDR6. Flex 140 places two small GPUs on a 75 W card, with separate 6 GB memories; these are not one coherent 12 GB GPU. Flex 170 uses one 32-core GPU with 16 GB and a 150 W board envelope. Shared technology with Arc explains the graphics and media engines, but server drivers, virtualization, cooling and supported workloads determine the datacenter product. A graphics-derived matrix engine does not imply GPU Max's FP64 or HBM characteristics.

Crescent Island is a distinct inference-oriented direction. The October 2025 OCP announcement described 160 GB LPDDR5X and customer sampling targeted for the second half of 2026. By Computex and Hot Chips 2026, Intel described up to 480 GB, a 350 W air-cooled PCIe card, 32 Xe3P cores and 256 XMX engines. The later capacity disclosure supersedes the earlier headline rather than proving that every future SKU has 480 GB. Sampling expectations are not general availability. No unsupported memory-bandwidth or low-precision throughput figure is inferred from the core count.

The architectural inference is that Intel is prioritizing resident model capacity, long contexts and installation in existing air-cooled facilities. Large memory can avoid model sharding and repeated host transfers, but capacity alone does not establish token latency. Decode may remain bandwidth-bound, while prefill demands matrix throughput; concurrency changes both. LPDDR and HBM therefore occupy different design points. The meaningful test is tokens per second at a specified latency, model, context length, precision and power limit, not whether one memory capacity number exceeds an HBM GPU's capacity.

Core Ultra extends the same heterogeneous-compute idea into the client and edge, combining CPU, integrated GPU and NPU. The NPU's low-power inference role differs from a datacenter training accelerator, while Arc-derived graphics and media technology connects to the Flex lineage. OpenVINO provides an inference deployment layer across supported CPU, GPU and NPU devices. Model conversion and common APIs help portability, but device plugins, supported operators, memory limits and precision still differ. Shared branding or a common runtime must not be used to transfer server GPU bandwidth or matrix rates to a client part.

16 Ethernet NICs, IPU and programmable networks

Network and infrastructure devices · 1/3

Product	GA / milestone	Status	Class	Ports × speed	SerDes	Host interface
E810	2019	Historical shipping	Ethernet NIC	Up to 100GbE; card-dependent	25G NRZ	PCIe 4
E830-CQDA2	2025	Shipping	Ethernet NIC	2 × 100GbE	50G PAM4	PCIe 5 x8 / PCIe 4 x16
E835	2026-06	Shipping	Ethernet NIC	1 × 200 / 2 × 100GbE; variants	n/d	n/d
IPU E2100-CCQDA2	2023-2024 documenta tion	Shipping	ASIC IPU	1 × 200 / 2 × 100GbE	50G PAM4 / 25G NRZ	PCIe 4 x16
Gaudi 3 integrated NIC	2024	Shipping	On-accelerator fabric	24 × 200GbE	48 × 112G PAM4	Internal to accelerator

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Network and infrastructure devices · 2/3

Product	Programmable engine	Embedded CPU	On-card memory	Transport / RDMA
E810	DDP pipeline profiles	n/a	n/d	iWARP / RoCEv2
E830-CQDA2	DDP / packet classification	n/a	n/d	Ethernet; RDMA SKU-dependent
E835	Configurable ports / packet processing	n/a	n/d	iWARP + RoCEv2
IPU E2100-CCQDA2	P4 pipeline	16 Arm Neoverse N1	48 GB LPDDR4x; 32 MB SLC	Ethernet; programmable storage transport
Gaudi 3 integrated NIC	Communication engines	n/a	Uses accelerator memory	RoCE

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Network and infrastructure devices · 3/3

Product	Congestion control	Key offloads	Power
E810	Protocol / driver-dependent	Virtualization, RSS, timing	n/d
E830-CQDA2	n/d	Packet processing, timing, telemetry	n/d
E835	n/d	RDMA, timing, security	n/d
IPU E2100-CCQDA2	Traffic shaping; implementation-specific	NVMe, IPsec, crypto, compression, isolation	n/d
Gaudi 3 integrated NIC	Ethernet fabric configuration	Device RDMA and collectives	Included in accelerator power

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Intel's conventional Ethernet controllers evolve from E810's 100 GbE generation toward E830 and E835 configurations reaching 200 GbE. These products support general server networking, virtualization, timing and selected RDMA capabilities. They should not be relabeled as dedicated AI collective engines merely because they can carry AI traffic. The host link, port allocation and transport configuration limit delivered bandwidth. A 200 Gb/s port is 25 GB/s in one direction before overhead; two connectors do not necessarily mean two independently available 200 Gb/s ports.

The IPU is a different architectural layer. Mount Evans/E2000 introduced an ASIC infrastructure processor co-developed with Google; E2100 adapter documentation specifies a 16-core Arm Neoverse N1 complex, a P4 packet pipeline, 48 GB LPDDR4x and hardware storage/security engines. The Arm cores run infrastructure software independently of the tenant host. The offload path can terminate virtual storage and networking while preserving a separate trust and management domain. E2000 denotes the family/architectural lineage; the E2100 adapter brief explicitly names an E2100 SoC, so those labels should not be collapsed into an unsupported silicon identity claim.

Google Cloud Next 2022 connected the IPU architecture to a real service: C3 combined fourth-generation Xeon with Google's custom Intel IPU, initially in private preview and generally available in May 2023. This is stronger deployment evidence than a generic roadmap slide. FPGA-based IPU and SmartNIC designs represent another implementation branch, trading reconfigurability for a different power, cost and software profile. Tofino's P4-programmable switch datapath is yet another class: switching and stateful packet processing do not make it a host IPU. Intel's discontinuation of the Tofino line limits its role to historical infrastructure and deployed systems.

Architecture Day 2021 made the implementation branches explicit. Oak Springs Canyon combined Xeon D and Agilex FPGA as an IPU reference platform, while Arrow Creek became the N6000 FPGA-based acceleration development platform/SmartNIC. Mount Evans used an ASIC approach. The distinction matters when evaluating programmability: FPGA logic can change the datapath itself, whereas an ASIC packet pipeline exposes a bounded programming model and general-purpose cores handle exceptions and control. None of these names alone establishes a particular AI collective-offload capability.

17 Interconnect, memory, optics and programmable logic

Separate bandwidth domains · 1/2

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
PCIe 3	2010s	8 GT/s	16	15.75
PCIe 4	Ice Lake / Gaudi 2	16 GT/s	16	31.51
PCIe 5	Sapphire / Gaudi 3	32 GT/s	16	63.02
CXL 1.1 → 2.0 → 3.0	SPR → Xeon 6 → Diamond	PCIe PHY generation-dependent	n/d	n/d
Gaudi 3 Ethernet	2024	112G PAM4 SerDes	2 per 200GbE	25
Xe Link / Max 1550	2023	n/d	n/d	n/d
Optical compute interconnect	2024 demo	32 Gb/s optical channel	n/a	250 aggregate TX, derived

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Separate bandwidth domains · 2/2

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
PCIe 3	Platform-dependent	Root / switch tree	n/a	I/O; no general cache coherence
PCIe 4	Platform-dependent	Root / switch tree	n/a	I/O DMA
PCIe 5	Platform-dependent	Root / switch tree	n/a	I/O; CXL is a separate protocol
CXL 1.1 → 2.0 → 3.0	Shared physical lane budget	Direct / switched by version	n/d	Cache / memory semantics by device type
Gaudi 3 Ethernet	24	3 links × 7 peers + 3 external	8 local	RDMA; not CPU cache coherence
Xe Link / Max 1550	n/d	System-wired GPU graph	4 GPU Lenovo reference	Peer GPU memory traffic
Optical compute interconnect	n/a	Optical chip-to-chip	n/d	Physical I/O demonstration, not a coherence protocol

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

UPI maintains CPU socket coherence, whereas PCIe is primarily an I/O transport and CXL adds defined cache and memory protocols over related physical links. Version support alone does not prove that every device type, switching configuration or pooling mode is enabled in a server. Real CXL measurements on Sapphire Rapids at MICRO 2023 show why latency, bandwidth asymmetry, placement and device implementation matter. A larger addressable memory tier is useful only if software places the right data there. The same reasoning applies to Optane: persistence requires crash-consistency mechanisms and has different read/write behavior from DRAM.

Intel's OFC 2024 optical compute interconnect demonstration co-packaged an optical I/O chiplet with a CPU and carried live data. The disclosed 4 Tb/s bidirectional aggregate is 500 GB/s combined, or 250 GB/s per direction under the stated symmetric convention. Its purpose is to extend reach and improve electrical-to-optical integration as electrical package escape becomes expensive. It is a technology demonstration, not evidence that shipping Xeons already expose an optical coherent fabric. Optical I/O, Ethernet switching and memory coherence solve different layers of the system problem.

Altera's Stratix and Agilex families bring reconfigurable datapaths, hardened interfaces and packaging experience into Intel's historical infrastructure portfolio. FPGA acceleration is attractive when a streaming pipeline, protocol or low-latency operation can be specialized; it is not interchangeable with a general training GPU. EMIB's role in FPGA integration also helps explain its later use in Xeon and GPU packages. Ownership changes the current portfolio boundary: the sale of 51% of Altera closed on 12 September 2025, leaving Intel with 49%. Treating Altera as a wholly owned Intel product group in 2026 would therefore be incorrect.

Optane was exited in 2022; a planned future persistent-memory module should not be counted as a shipped Sapphire Rapids feature. Omni-Path belongs to Intel's earlier HPC fabric history, while subsequent Cornelis products belong to that company's portfolio. Loihi 2 and Hala Point are neuromorphic research systems with event-driven computation and a different programming model, not drop-in datacenter GPUs. These technologies are important context, but none should be used to inflate the count of currently supported Intel AI training platforms.

18 Systems and software: the usable architecture

Representative deployed or documented systems · 1/2

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
AWS DL1	2021 GA	2 Xeon; 8 Gaudi	8 Gaudi
Aurora	2023 installation; later user service	2 Xeon Max + 6 GPU Max / node	6 GPU node; Xe Link topology
Lenovo SD650-I V3	2023; withdrawn 2024	Up to 2 Xeon + 4 GPU Max	4 GPU
Gaudi 3 HLB-325	2024	Host CPUs + 8 Gaudi 3	8 Gaudi
Google C3	2023 GA	Xeon 4th gen + custom IPU	CPU virtual machine

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Representative deployed or documented systems · 2/2

Platform	Scale-out per accelerator	Rack power	Cooling
AWS DL1	400Gb/s instance ÷ 8 = 6.25 GB/s average; shared	n/d	n/d
Aurora	8 x 200Gb/s NIC ÷ 6 = 33.3 GB/s average	n/d	Direct liquid
Lenovo SD650-I V3	Adapter-dependent	n/d	Direct water cooling
Gaudi 3 HLB-325	3 x 200GbE = 75 GB/s one-way	n/d	Air / liquid variants
Google C3	n/a	n/d	n/d

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Aurora’s two CPU Max and six GPU Max devices per node demonstrate why system ratios matter. Eight Slingshot endpoints provide scale-out connectivity, but dividing their aggregate rate by six is an average budget, not six dedicated identical NIC paths. Affinity among CPU NUMA domains, GPU tiles and network endpoints affects real communication. Likewise, an eight-Gaudi baseboard’s local fully connected graph is not an entire datacenter fabric. Oversubscription, collective scheduling and failure recovery determine cluster behavior beyond the node.

Software milestones and hardware relationship

Stack / release	Date / era	Hardware enabled	Key features / constraint
oneDNN / CPU framework libraries	2017-2026	Xeon AVX-512 / VNNI / AMX	ISA dispatch, packing and kernel selection; application must use enabled path
oneAPI / SYCL / Level Zero	2020s	Xe GPU / CPU	Shared programming model; placement and device-specific tuning remain
SynapseAI / Gaudi software 1.24	2026 documentation	Gaudi 2 / Gaudi 3	PyTorch integration, graph compilation, TPC kernels and HCCL
IPU SDK	2023-2024 documentation	E2100 IPU	P4 pipeline, infrastructure services and host integration
DSA / IAA software interfaces	2023 onward	Xeon integrated engines	Work queues, shared virtual addressing and asynchronous completion; overhead matters

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

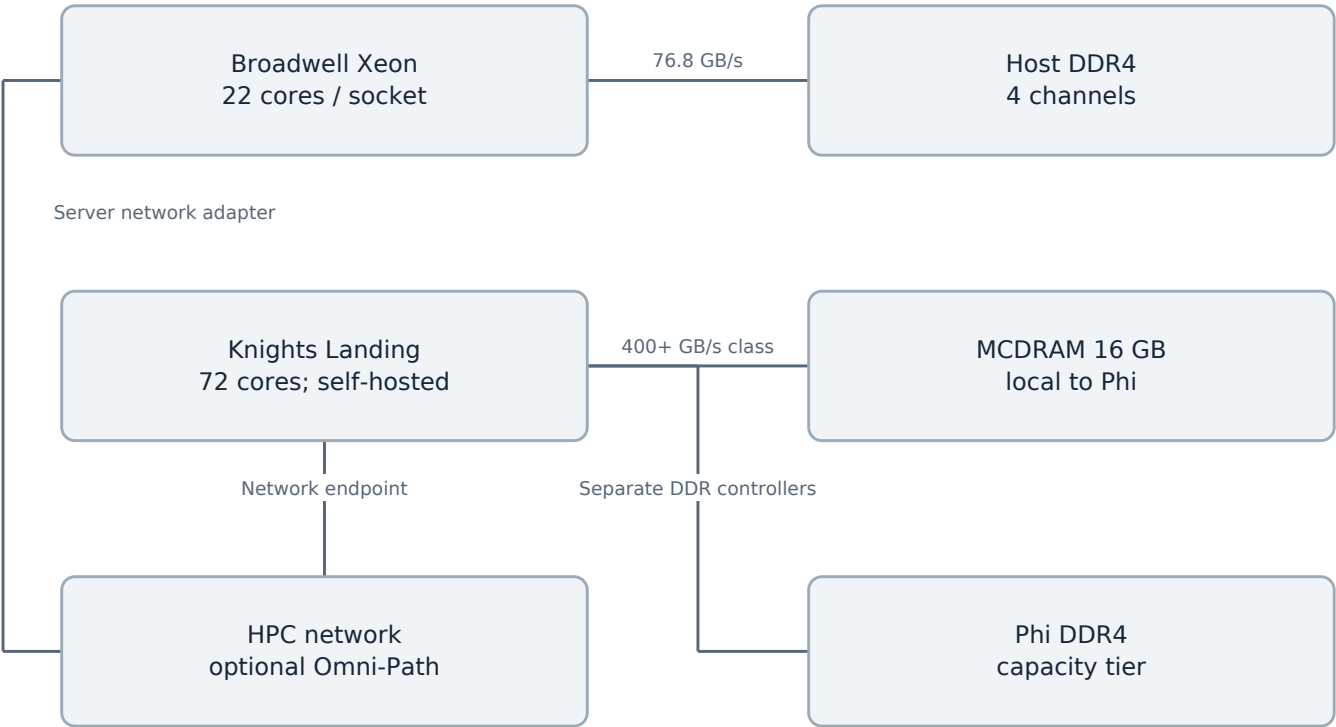
The software lines are related but not interchangeable. oneAPI/SYCL and Level Zero expose Intel GPU execution; Gaudi uses its own compiler/runtime and communication stack; Xeon libraries dispatch to vector or AMX kernels; IPU software manages infrastructure services. A common framework frontend does not guarantee equal operator coverage, numerical behavior or performance. Evaluate complete model graphs, fallback behavior, quantization, checkpoint portability and distributed communication. This is especially important when a hardware roadmap changes: a technically portable source program may still depend on a vendor-specific optimized kernel ecosystem.

OpenVINO is the inference-oriented bridge across Intel's CPU, GPU and NPU portfolio. It belongs alongside, rather than in place of, the lower-level compiler/runtime and mathematical libraries. For an enterprise inference deployment, supported model transformations, precision conversion, batching, device selection and profiling can matter more than the application seeing a particular instruction. Its hardware support matrix should be checked against the exact runtime release; a framework's generic device name is not a promise that every older Intel accelerator remains supported.

19 Four system views

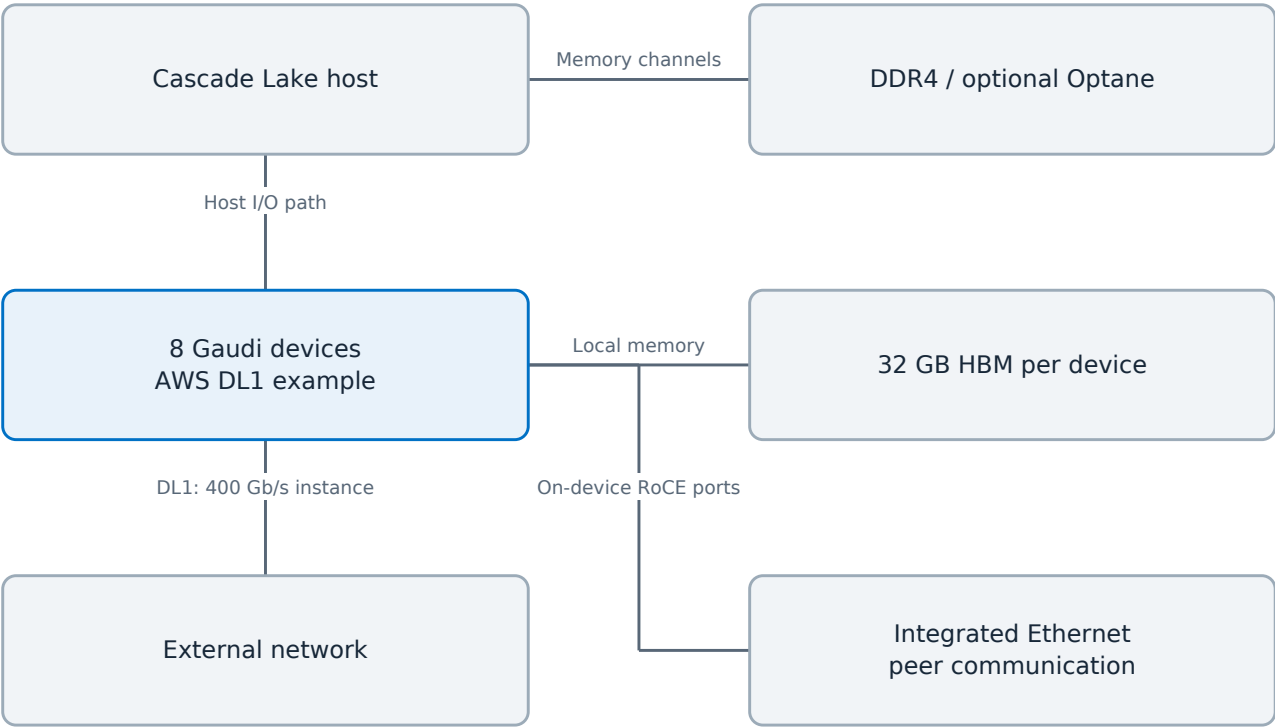
These diagrams summarize actual architectural relationships across four eras. Lines identify data paths, not necessarily coherent memory. A per-device bandwidth is not multiplied by device count unless an aggregate is explicitly requested. The drawings intentionally distinguish accelerator-local memory, host memory and external networking; none represents all memory as a uniform pool.

2015-2016: host CPU and self-hosted many-core



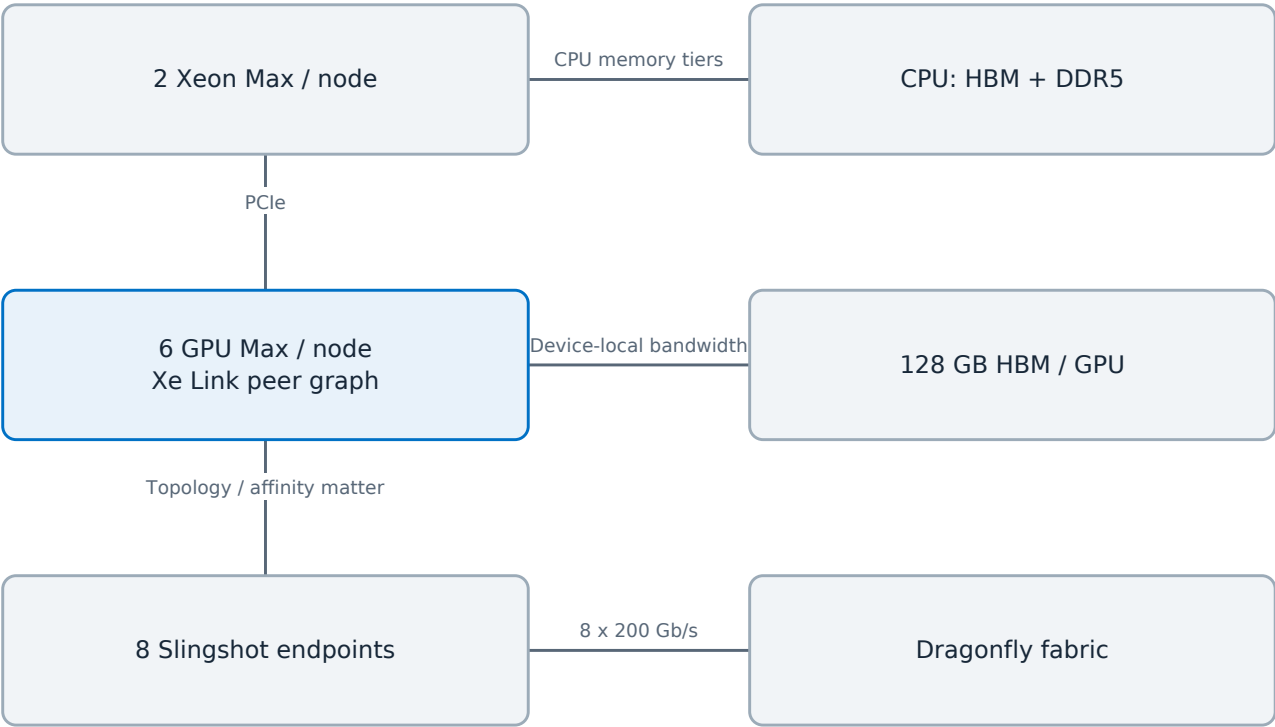
Xeon and Phi are alternative node constructions here; the diagram does not imply Knights Landing is attached to Broadwell as a mandatory coprocessor.

2019-2021: CPU plus Ethernet-connected AI devices



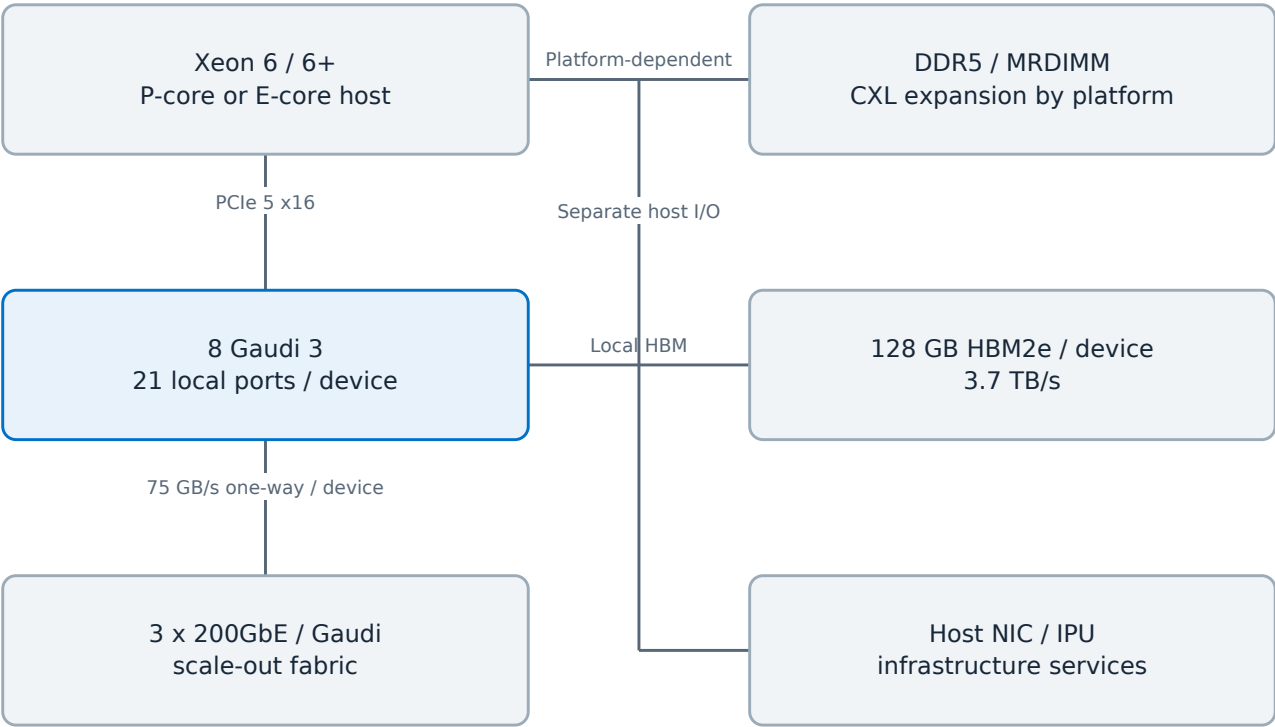
DL1’s external service bandwidth differs from Gaudi’s on-chip aggregate network port bandwidth. Optane is a CPU platform option, not Gaudi HBM.

2023-2024: Aurora’s heterogeneous memory hierarchy



An average 33.3 GB/s scale-out budget per GPU follows from 8 × 200 Gb/s divided by six; it is not a uniform dedicated link allocation.

2024-2026: specialized compute and infrastructure paths



This is a role diagram, not a claim of one mandatory OEM configuration. Crescent Island is a separate announced PCIe inference branch with LPDDR5X; Diamond Rapids is a disclosed future host platform.

20 Normalized ratios and design implications

Derived ratios from stated reference configurations

Reference	DRAM GB/s per core	L3 MB per core	Interpretation
Broadwell 2699 v4	3.49	2.50	22 cores share four channels
Skylake 8180	4.57	1.375	L2 grows; L3 changes inclusion policy
Sapphire 8490H	5.12	1.875	AMX raises arithmetic demand
Emerald 8592+	5.60	5.00	Large LLC capacity increase
Sierra 6780E	2.84	0.75	Density-oriented reference
Granite 6980P / MRDIMM	6.60	3.9375	12 channels at 8800 MT/s
Clearwater 6990E+	2.67	2.00	Double cores; more cache per core

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Accelerator balance, dense BF16 basis

Device	HBM bytes/FLOP	All-port network / HBM BW	External BW / device	BF16 peak / device power
Gaudi 2	0.00569	0.122	37.5 GB/s, 3 x 100GbE allocation	0.720 TF/W at 600 W
Gaudi 3	0.00221	0.162	75 GB/s, 3 x 200GbE allocation	1.864 TF/W at 900 W
GPU Max 1550	0.00381	0.0994 Xe Link / HBM, Lenovo basis	System-dependent	1.398 TF/W at 600 W

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Gaudi 3 increases matrix peak more quickly than HBM bandwidth relative to Gaudi 2: the HBM bytes available per peak BF16 operation fall by more than half on the selected whitepaper basis. More reuse, larger useful tiles or greater batching is needed to approach compute peak. Network bandwidth also remains a fraction of local HBM bandwidth. Communication-heavy parallelism cannot be judged by arithmetic scaling alone. These ratios describe resource balance; they do not predict a model benchmark without its arithmetic intensity and communication pattern.

Peak FLOPS divided by board or device power is an upper-level arithmetic normalization, not measured energy efficiency. It omits host CPUs, DIMMs, switches, optics, cooling and utilization. No cross-vendor performance-per-watt ranking is inferred from this table. For CPUs, per-core memory bandwidth is similarly an equal-share budget, not a guaranteed allocation. The useful engineering question is which layer saturates first and whether software can increase locality or overlap before more hardware is added.

21 What the major architecture conferences add

Hot Chips and ISSCC answer complementary questions. Hot Chips usually exposes the block diagram, hierarchy, interfaces and product intent. ISSCC can explain the physical implementation that makes those blocks practical: multi-die signaling, energy per bit, package construction, clocking and power. The strongest examples here are Sapphire Rapids' quasi-monolithic EMIB fabric, Ponte Vecchio's multi-process 3D integration and Emerald Rapids' cache-rich two-tile design. Conference numbers must remain attached to their prototype, measurement condition and disclosure date before being reconciled with commercial SKU data.

ISCA, MICRO, HPCA and ASPLOS are particularly valuable for testing how product mechanisms behave. LIA at ISCA 2025 uses AMX-enabled Xeons and a GPU cooperatively, showing that CPU matrix execution can change offloading decisions. MICRO 2023's real-device CXL study distinguishes actual memory expansion from software emulation. ASPLOS 2024's DSA analysis identifies when asynchronous data movement pays for its submission and completion overhead. HPCA 2020's persistent-memory keynote examines the programming consequences of capacity, persistence and asymmetric access. These are research or measurement results, not proof that Intel adopted every proposed mechanism in silicon.

The conference index distinguishes direct product disclosure, implementation disclosure, independent measurement, and deployment evidence. DAC and Foundry Direct Connect are useful for design enablement, packaging and EDA context; MWC emphasizes telecom and edge; OCP emphasizes platform integration; Computex and Intel's own launch events establish product positioning and availability. Cloud Next and AWS disclosures connect chips to services. No specific GTC, Microsoft Ignite or DAC product-architecture talk is asserted where a qualifying primary record was not established in this review. That is a bounded coverage gap, not evidence that Intel never participated.

Conference-to-product index - 1/2

Venue	Year	Paper / talk / disclosure	Presenter / authors
Hot Chips	2015	Knights Landing	Avinash Sodani
Hot Chips	2018	Cascade Lake	Akhilesh Kumar
Hot Chips	2019	Nervana; packaging; Optane	Intel
HPCA	2020	Persistent Memory and the Path to Being Comfortably NUMB	Steven Swanson
ISSCC	2022	Ponte Vecchio: A Multi-Tile 3D Stacked Processor for Exascale Computing	W. Gomes et al.
ISSCC	2022	Sapphire Rapids: Next-Generation Xeon	Intel
Google Cloud Next	2022	C3 and custom Intel IPU	Google / Intel
Hot Chips	2023	Granite Rapids and Sierra Forest	Intel
MICRO	2023	Demystifying CXL Memory with Genuine CXL-Ready Systems and Devices	Yan Sun et al.
ISSCC	2024	Emerald Rapids: 5th-Generation Xeon Scalable	Ashley O. Munch et al.
ASPLOS	2024	A Quantitative Analysis and Guidelines of Data Streaming Accelerator in Modern Intel Xeon Scalable Processors	Reese Kuper et al.
Hot Chips	2024	Gaudi 3; Xeon 6; optical I/O	Intel
OFC	2024	Integrated optical compute interconnect	Intel IPS
Intel Vision	2024	Gaudi 3 architecture and open systems	Intel

Venue	Year	Paper / talk / disclosure	Presenter / authors
ISCA	2025	LIA: cooperative AMX CPU-GPU computation and CXL offloading	Hyungyo Kim et al.
Hot Chips / Tech Tour	2025	Clearwater Forest	Intel
OCP Global Summit	2025	Crescent Island initial disclosure	Intel
MWC	2026	Xeon 6+ network and edge positioning	Intel
Computex	2026	Xeon 6+ and E835 launch	Intel
Hot Chips	2026	Diamond Rapids and Crescent Island	Intel
Foundry Direct Connect	2024-2025	Process and packaging enablement	Intel / EDA partners

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Conference-to-product index - 2/2

Venue	Product	Evidence type	Source
Hot Chips	Xeon Phi	Product architecture	C13
Hot Chips	Xeon 2nd gen	Architecture / VNNI / memory	C06
Hot Chips	NNP-T / NNP-I	Official press release	C14
HPCA	Optane context	Independent research keynote	R05
ISSCC	Ponte Vecchio	Implementation; paper 2.1 press kit	C03
ISSCC	Sapphire Rapids	Implementation; paper 2.2 press kit	C03
Google Cloud Next	Sapphire Rapids + IPU	Preview; GA followed 2023	E15
Hot Chips	Xeon 6	Architecture briefing slides	C12
MICRO	Sapphire Rapids / CXL	Real-device measurement	R04
ISSCC	Emerald Rapids	Implementation; paper 2.3 press kit	C04
ASPLOS	DSA / Sapphire Rapids	Measured offload behavior	R02
Hot Chips	Gaudi 3 / Xeon 6 / OCI	Architecture and demo summary	E13
OFC	OCI chiplet	Live technology demonstration	E12
Intel Vision	Gaudi 3	Product announcement	E20
ISCA	Sapphire / Granite Rapids	Independent system research	R01
Hot Chips / Tech Tour	Xeon 6+	Architecture / packaging	P08
OCP Global Summit	Crescent Island	Roadmap / sampling target	E03

Venue	Product	Evidence type	Source
MWC	Clearwater Forest	Platform demonstration	E16
Computex	Clearwater / E835 / Crescent	Commercial launch / update	E02
Hot Chips	Diamond / Crescent	Architecture announcement	E01
Foundry Direct Connect	18A / EMIB / Foveros	Technology ecosystem	E19

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

22 Architecture conclusions and remaining gaps

Intel's durable architectural assets are the CPU software base, increasingly specialized host execution, heterogeneous packaging, and infrastructure offload. Its accelerator history is more fragmented. Gaudi offers a distinctive Ethernet-centered system; GPU Max demonstrates ambitious HPC integration; Crescent Island pursues capacity-oriented inference economics. These are complementary design points, not interchangeable generations of one GPU. The correct unit of comparison is the deployable workload system: compute format, memory residence, locality, fabric, power envelope and maintained software path.

The remaining material gaps are concentrated in final Diamond Rapids SKU and availability details, Crescent Island's memory bandwidth and arithmetic rates, the precise shipping configuration matrix of newer NICs, and restricted conference slide or full-paper access. The report therefore uses official announcement-level claims for those products and leaves unsupported fields as n/d. Earlier packaging roadmaps are reconciled with later launch records rather than treated as actual shipment dates. The canonical data file preserves source IDs, claim scope, numerical basis and product status so these gaps can be updated without changing the report structure.

Source map

- 01 Executive synthesis and scope - C03, C12, E01, E02, E07, E11, E12, E17, P02, P03, P05, P06, P10, P14
- 02 How to read the numbers - P02, P05, P09, P16, P17
- 03 Xeon generation atlas - C03, C04, C06, C12, E01, E02, E16, P01, P03, P05, P06, P08, P13, P16, P17, P18, P19, P20, P21, P22, P23, P24, P25, P26, P35
- 04 Haswell and Broadwell: the monolithic baseline - P05, P07, P20
- 05 Skylake, Cascade Lake and Cooper Lake - C06, P05, P06, P22, P23
- 06 Ice Lake: a wider core and a wider server - P05, P07, P24
- 07 Sapphire Rapids and Xeon Max: tiles and engines - C03, P03, P05, P13, R01, R02
- 08 Emerald Rapids: cache and integration efficiency - C04, P03, P26
- 09 Xeon 6: separating performance and density - C12, E04, E06, P16, P17, P18
- 10 Clearwater Forest and Diamond Rapids - E01, E02, E08, E16, P01, P08, P17, P19, P35
- 11 Accelerator generation atlas - C03, C13, E01, E03, E05, E14, P02, P05, P09, P11, P27, P28, P29, P30, P37
- 12 Xeon Phi and Nervana: earlier parallel-compute paths - C13, C14, E17, P05, P28, P30
- 13 Gaudi 1, 2 and 3: Ethernet inside the accelerator - E09, E14, P02, P04, P09, P30
- 14 Ponte Vecchio: a heterogeneous HPC package - C03, E17, P11, P27, P31
- 15 Flex, Arc and Crescent Island: inference branches - E01, E02, E03, P11, P29, P38, R01
- 16 Ethernet NICs, IPU and programmable networks - E02, E15, E18, E21, P02, P10, P14, P15, P32
- 17 Interconnect, memory, optics and programmable logic - C12, C14, E01, E07, E10, E11, E12, E13, E17, P02, P03, P05, P24, P27, P33, R04, R05
- 18 Systems and software: the usable architecture - E14, E15, P02, P03, P05, P10, P11, P27, P30, P31, P34, P38, R02
- 19 Four system views - E01, P02, P05, P31
- 20 Normalized ratios and design implications - P01, P02, P03, P06, P16, P17, P18, P19, P20, P21, P25, P26, P27, P30
- 21 What the major architecture conferences add - C03, C04, C06, C12, C13, C14, E01, E02, E03, E12, E13, E14, E15, E16, E19, E20, P08, R01, R02, R04, R05
- 22 Architecture conclusions and remaining gaps - E01, E02, E03, E07, P02, P03, P10, P14

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